

SOCIAL MEDIA USAGE ANALYSIS USING POWER BI DASHBOARD

1. Project Overview and Objective:

This project, titled "Social Media Usage Analysis using Power BI Dashboard", explores the complex relationships between individuals' personal demographics, lifestyle factors, and their Instagram usage patterns. It leverages a comprehensive dataset of 5,000 users, originally from an Instagram usage survey, which has been thoroughly cleaned, enriched with derived features, and segmented into three focused Excel files: `Personal_Cleaned_data.xlsx`, `Lifestyle_Cleaned_data.xlsx`, and `SocialMedia_Cleaned_data.xlsx`.

The analysis integrates key dimensions:

- **Personal/Demographic** data (age, gender, country, urban/rural location, income level, employment status, education, relationship status, and derived flags like `Age_Group`, `Is_Urban`, `High_income`).
- **Lifestyle/Health** indicators (sleep hours, perceived stress, happiness, BMI, blood pressure, work hours, `Bodyweight_class`, `Sleep_Category`, `BP_Category`, `Stress_Happiness_Index`, `Work_Life_Balance_Score`, `High_Risk_Flag`).
- **Social Media Behavior** metrics (daily active minutes on Instagram, sessions per day, ads viewed, followers/following, engagement scores, content preferences, reels/stories consumption, `Usage_Group`, `Engagement_class`, `High_Engagement_Flag`, etc.).

Data preparation involved extensive cleaning (handling inconsistencies, missing values, and outliers), feature engineering (e.g., categorizations, ratios like `Follower_Ratio`, composite indices, and flags), and integration across tables via the common `user_id` key. The processed dataset was imported into Power BI, where a robust data model was built with relationships, calculated columns, and DAX measures. Interactive visualizations (charts, slicers, maps, KPIs, etc.) were developed to uncover actionable insights.

The resulting Power BI dashboard enables dynamic exploration of how social media usage correlates with (or impacts) users' well-being, productivity, and demographics—highlighting trends such as high-engagement users' lifestyle risks, regional differences, and behavioral segments.

- **Understand Usage Patterns:** Identify distinct user segments based on Instagram activity levels (e.g., Low/Moderate/Heavy Usage, `Activity_Level`) and engagement (`High_Engagement_Flag`, `user_engagement_score`) to reveal consumption behaviors like reels/stories viewed, session lengths, and content preferences (Tech, Fashion, Food, etc.).
- **Analyze Correlations with Lifestyle & Health:** Examine relationships between social media metrics and health indicators—such as sleep quality, stress/happiness levels, BMI, blood pressure, and work-life balance—to assess potential impacts (e.g., heavy usage linked to higher stress or poorer sleep).
- **Demographic & Socio-Economic Insights:** Investigate how factors like age group, gender, income, employment, urban/rural setting, and country influence Instagram usage, engagement, and associated lifestyle outcomes.
- **Risk & Well-Being Assessment:** Highlight high-risk profiles (e.g., `High_Risk_Flag` users with hypertension or poor metrics) and their social media behaviors, supporting recommendations for healthier digital habits.
- **Interactive Visualization & Decision Support:** Deliver a user-friendly Power BI dashboard with filters, drill-downs, and KPIs for stakeholders (researchers, marketers, public health analysts) to explore data interactively and derive data-driven conclusions.

2. Data Source:

The dataset used in this project was obtained from Kaggle, a leading platform for data science and machine learning datasets.

- Platform: Kaggle
- Domain: Social Media
- Dataset Title: Social Media User Analysis (Instagram Usage & Lifestyle dataset)

This is a synthetic yet realistic dataset simulating Instagram user behavior combined with demographic, lifestyle, and health-related attributes. The original dataset contains over 1 million user records, but a representative subset of 5,000 users was used for this project.

It includes comprehensive information across three main categories:

- Personal/Demographic details (age, gender, country, income level, employment status, education, relationship status, etc.)
- Lifestyle & Health indicators (sleep hours, perceived stress score, self-reported happiness, BMI, blood pressure, weekly work hours, etc.)
- Social Media Usage metrics specific to Instagram (daily active minutes, sessions per day, followers/following count, engagement score, content preferences, reels/stories viewed, subscription status, etc.)

Key Characteristics

- Number of Records: 5,000 users
- Original File Format: CSV (processed into multiple cleaned Excel files for this project)
- Total Features: Approximately 50+ columns (after cleaning and feature engineering)
- Data Type: Synthetic data designed to mimic real-world Instagram usage patterns in 2025–2026
- License: Typically available under Kaggle’s standard open usage for educational and analytical purposes

Data Acquisition & Preparation

The raw dataset was downloaded from Kaggle and underwent extensive data cleaning, transformation, and feature engineering processes. The data was then divided into three separate Excel files for better manageability:

- Personal_Cleaned_data.xlsx
- Lifestyle_Cleaned_data.xlsx
- SocialMedia_Cleaned_data.xlsx

These cleaned files were imported into Power BI for modeling, relationship building, and dashboard development.

This dataset is ideal for analyzing the impact of social media usage on lifestyle, mental well-being, and demographic trends.

3. Problem Statement:

In today's digital era, social media platforms, particularly Instagram, have become an integral part of daily life. While these platforms offer connectivity, entertainment, and information, excessive or uncontrolled usage has raised significant concerns regarding its impact on individuals' mental and physical well-being. Many users experience disrupted sleep patterns, elevated stress levels, reduced happiness, poor work-life balance, and increased health risks such as hypertension and abnormal BMI — all potentially linked to heavy social media consumption.

Despite the growing volume of social media data, there is limited actionable insight into how Instagram usage patterns (daily active time, session frequency, content engagement, reels/stories consumption, etc.) correlate with users' demographic profiles and lifestyle/health indicators. Most existing studies focus on either usage statistics or health outcomes in isolation, lacking an integrated view that connects personal characteristics, lifestyle habits, and social media behavior.

Core Problem

How does Instagram usage intensity and behavior influence users' lifestyle quality, mental well-being, and health risks across different demographic segments?

This lack of comprehensive analysis makes it difficult for individuals, healthcare professionals, policymakers, and digital platform designers to:

- Identify high-risk user groups who may be negatively affected by social media.
- Understand demographic and socio-economic factors driving problematic usage.
- Develop targeted interventions or features to promote healthier digital habits.

4. Attribute Details:

a. Personal_Cleaned_data

Attribute Name	Data Type	Description
user_id	Integer	Unique identifier for each respondent
age	Integer	Age of the respondent in years
gender	String	Self-reported gender (e.g., Male, Female, Non-binary, Prefer not to say)
Gender2	String	Standardized/cleaned gender category
country	String	Country of residence
Country2	String	Standardized/cleaned country name
urban_rural	String	Location type (Urban, Rural, Suburban)
income_level	String	Income bracket (Low, Lower-middle, Middle, Upper-middle, High)
employment_status	String	Employment status (e.g., Full-time employed, Student, Retired, Unemployed)
education_level	String	Highest education level (e.g., High school, Bachelor, Master, PhD)
Employment_Status2	String	Standardized/cleaned employment status
relationship_status	String	Relationship status (e.g., Single, Married, In a relationship, Divorced)
Age_Group	String	Age group category (e.g., Teen, 18-24, 25-34, 35-44, 45-54, 55+)
Is_Urban	Integer	Binary flag: 1 if urban, 0 otherwise
Is_female	Integer	Binary flag: 1 if female, 0 otherwise
Is_employed	Integer	Binary flag: 1 if employed, 0 otherwise
Is_single	Integer	Binary flag: 1 if single, 0 otherwise
High_income	Integer	Binary flag: 1 if high income, 0 otherwise
Income_score	Integer	Numeric score derived from income level (1-5 scale)
Age_Numeric_group	Float	Numeric midpoint of the age group for analysis

b. Lifestyle_Cleaned_data

Attribute Name	Data Type	Description
user_id	Integer	Unique identifier for each individual in the dataset
sleep_hours_per_night	Float	Average hours of sleep per night reported by the individual
perceived_stress_score	Integer	Self-reported stress level score (higher values indicate greater stress)
self_reported_happiness	Integer	Self-reported happiness level score (higher values indicate greater happiness)
body_mass_index	Float	Body Mass Index (BMI) value calculated from height and weight
blood_pressure_systolic	Integer	Systolic blood pressure reading (mmHg)
blood_pressure_diastolic	Integer	Diastolic blood pressure reading (mmHg)
weekly_work_hours	Float	Average number of hours worked per week
Bodyweight_class	String	Categorical classification of body weight (e.g., Normal, Overweight, Obese, Underweight)
Sleep_Category	String	Categorical assessment of sleep quality (e.g., Poor, Good, Excellent)
BP_Category	String	Blood pressure category (e.g., Hypertension)
Stress_Happiness_Index	Float	Derived index combining stress and happiness scores
Work_Life_Balance_Score	Float	Calculated score representing work-life balance
High_Risk_Flag	String	Flag indicating high health/lifestyle risk status (e.g., High Risk)

c. SocialMedia_Cleaned_data

Attribute Name	Data Type	Description
user_id	Integer	Unique identifier for each user in the dataset
social_events_per_month	Integer	Number of social events attended per month
books_read_per_year	Integer	Number of books read by the user per year
daily_active_minutes_instagram	Integer	Average daily minutes spent active on Instagram
sessions_per_day	Integer	Number of app sessions per day
ads_viewed_per_day	Integer	Number of advertisements viewed per day
followers_count	Integer	Total number of followers on the platform
following_count	Integer	Total number of accounts the user is following
account_creation_year	Integer	Year the social media account was created
last_login_date	Integer	Numeric representation of the last login date
average_session_length_minutes	Float	Average length of each session in minutes
preferred_content_theme	String	User's preferred content category/theme (e.g., Tech, Fashion, Food)
privacy_setting_level	String	Privacy setting of the account (e.g., Public, Private, Friends only)
subscription_status	String	Current subscription tier/status
Subscription_status2	String	Alternative/duplicate subscription status field
user_engagement_score	Float	Overall engagement score for the user
posts_created_per_week	Integer	Average number of posts created per week
reels_watched_per_day	Integer	Number of reels watched per day
stories_viewed_per_day	Integer	Number of stories viewed per day
Usage_Group	String	Categorical group based on usage intensity (e.g., Low Usage, Moderate Usage, Heavy Usage)
Engagement_class	String	Classification of user engagement level (e.g., Low, Medium, High)
Follower_Ratio	Float	Ratio of followers to following count
Total_Content_Consumed_Daily	Integer	Total daily content consumption (reels + stories + etc.)
Engagement_Score_Adjusted	Float	Adjusted engagement score (possibly normalized)
High_Engagement_Flag	String	Flag indicating high engagement status
Activity_Level	String	Overall activity level category
Books_per_Social_Event	Float	Derived metric: books read per social event

5. Tools & Technologies:

- Microsoft Excel: Used for data cleaning, data transformation, feature engineering, creating new calculated columns, Pivot Tables for initial analysis, and splitting the original dataset into three cleaned files (Personal_Cleaned_data.xlsx, Lifestyle_Cleaned_data.xlsx, and SocialMedia_Cleaned_data.xlsx).
- Power BI Desktop: Primary tool for data modeling (establishing relationships between tables), writing DAX measures and calculated columns, creating interactive visualizations, and designing the complete dashboard.
- Power Query Editor (in Power BI): Utilized for advanced data import, cleaning, and transformation processes.

6. Data Pre-Processing (Excel/Query):

Data pre-processing was an essential and time-consuming stage in this project. The raw dataset obtained from Kaggle contained a large number of records and various attributes related to user demographics, lifestyle factors, and Instagram usage behavior. Before any meaningful analysis or visualization could be performed, the data needed to be carefully examined and prepared.

The pre-processing work started in Microsoft Excel. First, the original dataset was loaded and thoroughly inspected to identify issues such as missing values, duplicate records, inconsistent formatting, and incorrect data types. Cleaning operations were carried out systematically. This included removing or filling missing values where appropriate, correcting spelling and formatting inconsistencies in categorical fields like country names, education levels, employment status, and gender categories. Data standardization was also performed to ensure uniformity across the dataset.

A major part of the pre-processing involved feature engineering. Several new columns were created to extract more value from the existing data. These included age groups, body weight categories based on BMI, sleep quality categories, blood pressure categories, and a high-risk health flag. Additional calculated fields were developed such as stress-happiness index, work-life balance score, follower ratio, total daily content consumed, and various binary flags to simplify analysis. These engineered features helped in better understanding the relationships between social media usage and lifestyle outcomes.

To make the dataset more manageable and organized, the original file was divided into three separate cleaned Excel files: `Personal_Cleaned_data.xlsx` focusing on demographic information, `Lifestyle_Cleaned_data.xlsx` containing health and lifestyle related attributes, and `SocialMedia_Cleaned_data.xlsx` covering all Instagram usage metrics and engagement details.

Once the Excel files were ready, they were imported into Power BI using Power Query Editor. In Power Query, further transformations were applied such as changing data types, removing irrelevant columns, and merging the three tables using the common `user_id` column. Relationships were established between the tables to create a unified data model. This step ensured that the data was clean, consistent, and properly structured for building measures, calculations, and visualizations.

Overall, this detailed pre-processing phase transformed the raw dataset into a high-quality, analysis-ready format that enabled accurate insights and an effective interactive dashboard in Power BI.

7. Data Modeling & DAX

Data modelling is a critical step in Power BI that defines how different tables are connected and how information flows between them for accurate analysis and reporting. In this project, a well-structured relational data model was built using the three cleaned Excel files.

The data model consists of three primary tables:

- `Personal_Cleaned_data` – contains demographic and personal attributes such as age, gender, country, income level, employment status, education, relationship status, and various derived flags.
- `Lifestyle_Cleaned_data` – includes health and lifestyle-related information like sleep hours, stress score, happiness level, BMI, blood pressure, work hours, body weight class, sleep category, BP category, high risk flag, and work-life balance score.

- `SocialMedia_Cleaned_data` – holds all Instagram usage metrics such as daily active minutes, sessions per day, followers count, engagement score, content preferences, reels and stories viewed, usage group, and engagement class.

These three tables are linked through the common `user_id` column, which acts as the primary key. A one-to-one relationship was established between the tables because each `user_id` appears only once across all datasets. This creates a clean, unified structure where any field from one table can be combined with fields from the other tables for analysis.

The model follows a simple yet effective design that allows seamless integration of personal characteristics, lifestyle factors, and social media behavior in a single report. This relational setup eliminates data redundancy and ensures that filters applied on one table (for example, age group or country) automatically affect visualizations across lifestyle and social media metrics.

In addition to the data model, several DAX functions and measures were created in Power BI to support deeper analysis. These include:

- Calculated columns for categorizing users and deriving new insights.
- Measures for calculating averages, percentages, and ratios (such as average daily active minutes, engagement rate, or percentage of high-risk users).
- Conditional measures to dynamically show metrics based on slicer selections.
- KPI-related calculations for tracking high engagement users and health risk indicators.

The combination of a solid relational data model and custom DAX calculations enabled the creation of dynamic, interactive visualizations and allowed for complex cross-table analysis. This foundation was key to uncovering meaningful relationships between Instagram usage patterns and users' lifestyle and health outcomes.

Screenshot of Data Modeling

NameLifestyle

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

StructureRelationshipsCalculationsCalendars

	weekly_work_hours	Bodyweight_class	Sleep_Category	BP_Category	Stress_Happiness_Index	Work_Life_Balance_Score	High_Risk_Flag
72	38.5	Normal	Good	Hypertension	2.3	1.06944444444444	High Risk
75	37.7	Normal	Good	Hypertension	2.5	0.86666666666667	High Risk
80	34.7	Normal	Good	Hypertension	0.7	0.825	High Risk
89	55.4	Normal	Good	Hypertension	1.2	0.707865168539326	High Risk
77	34.4	Normal	Good	Hypertension	2.6	0.922077922077922	High Risk
60	51.5	Normal	Good	Hypertension	0.9	1.11666666666667	High Risk
97	20.8	Normal	Good	Hypertension	2.2	0.618556701030928	High Risk
88	36.4	Normal	Good	Hypertension	1.9	0.784090909090909	High Risk
61	54.5	Normal	Good	Hypertension	1.5	1	High Risk
73	46	Normal	Good	Hypertension	2.6	0.863013698630137	High Risk
63	26.3	Normal	Good	Hypertension	0.7	0.984126984126984	High Risk
94	22.4	Normal	Good	Hypertension	0.9	0.723404255319149	High Risk
81	26.5	Normal	Good	Hypertension	2.6	0.938271604938272	High Risk
79	47.4	Normal	Good	Hypertension	0.4	0.860759493670886	High Risk
70	44.6	Normal	Good	Hypertension	3	1.05714285714286	High Risk
92	43.7	Normal	Good	Hypertension	1.3	0.858695652173913	High Risk
81	24.3	Normal	Good	Hypertension	1.7	0.790123456790123	High Risk
78	32.3	Normal	Good	Hypertension	1.2	0.833333333333333	High Risk
62	31.6	Normal	Good	Hypertension	0.4	1.14516129032258	High Risk
93	41.6	Normal	Good	Hypertension	3.6	0.720430107526882	High Risk

Data

Search

Lifestyle

Avg Happiness

Avg Sleep Hours

Avg Stress Score

blood_pressure_diastolic

blood_pressure_systolic

body_mass_index

Bodyweight_class

BP_Category

High Risk %

High_Risk_Flag

perceived_stress_score

self_reported_happiness

Sleep_Category

sleep_hours_per_night

Stress_Happiness_Index

user_id

NameLifestyle

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

StructureRelationshipsCalculationsCalendars

	weekly_work_hours	Bodyweight_class	Sleep_Category	BP_Category	Stress_Happiness_Index	Work_Life_Balance_Score	High_Risk_Flag
72	38.5	Normal	Good	Hypertension	2.3	1.06944444444444	High Risk
75	37.7	Normal	Good	Hypertension	2.5	0.86666666666667	High Risk
80	34.7	Normal	Good	Hypertension	0.7	0.825	High Risk
89	55.4	Normal	Good	Hypertension	1.2	0.707865168539326	High Risk
77	34.4	Normal	Good	Hypertension	2.6	0.922077922077922	High Risk
60	51.5	Normal	Good	Hypertension	0.9	1.11666666666667	High Risk
97	20.8	Normal	Good	Hypertension	2.2	0.618556701030928	High Risk
88	36.4	Normal	Good	Hypertension	1.9	0.784090909090909	High Risk
61	54.5	Normal	Good	Hypertension	1.5	1	High Risk
73	46	Normal	Good	Hypertension	2.6	0.863013698630137	High Risk
63	26.3	Normal	Good	Hypertension	0.7	0.984126984126984	High Risk
94	22.4	Normal	Good	Hypertension	0.9	0.723404255319149	High Risk
81	26.5	Normal	Good	Hypertension	2.6	0.938271604938272	High Risk
79	47.4	Normal	Good	Hypertension	0.4	0.860759493670886	High Risk
70	44.6	Normal	Good	Hypertension	3	1.05714285714286	High Risk
92	43.7	Normal	Good	Hypertension	1.3	0.858695652173913	High Risk
81	24.3	Normal	Good	Hypertension	1.7	0.790123456790123	High Risk
78	32.3	Normal	Good	Hypertension	1.2	0.833333333333333	High Risk
62	31.6	Normal	Good	Hypertension	0.4	1.14516129032258	High Risk
93	41.6	Normal	Good	Hypertension	3.6	0.720430107526882	High Risk

Data

Search

Lifestyle

Avg Happiness

Avg Sleep Hours

Avg Stress Score

blood_pressure_diastolic

blood_pressure_systolic

body_mass_index

Bodyweight_class

BP_Category

High Risk %

High_Risk_Flag

perceived_stress_score

self_reported_happiness

Sleep_Category

sleep_hours_per_night

Stress_Happiness_Index

user_id

Name

Personal

Manage relationships

New measure

Quick measure

New column

New table

Mark as date table

StructureRelationshipsCalculationsCalendars

	user_id	age	gender	Gender2	country	Country2	urban_rural	income_level	employment_status	education_level	Employment
48	21	Male	Male	United States	United States	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er	
109	21	Male	Male	Brazil	Brazil	Urban	Low	Full-time employed	Bachelor	Full-time er	
120	47	Male	Male	South Korea	South Korea	Urban	Low	Full-time employed	Bachelor	Full-time er	
151	38	Male	Male	United Kingdom	United Kingdom	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er	
161	63	Male	Male	Australia	Australia	Urban	Low	Full-time employed	Bachelor	Full-time er	
253	50	Male	Male	Australia	Australia	Urban	Low	Full-time employed	Bachelor	Full-time er	
263	45	Male	Male	United States	United States	Urban	Middle	Full-time employed	Bachelor	Full-time er	
481	59	Male	Male	Germany	Germany	Urban	Middle	Full-time employed	Bachelor	Full-time er	
635	61	Male	Male	United Kingdom	United Kingdom	Urban	Middle	Full-time employed	Bachelor	Full-time er	
694	56	Male	Male	India	India	Urban	Middle	Full-time employed	Bachelor	Full-time er	
706	17	Male	Male	India	India	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er	
891	42	Male	Male	India	India	Urban	Middle	Full-time employed	Bachelor	Full-time er	
916	50	Male	Male	South Korea	South Korea	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er	
937	14	Male	Male	United States	United States	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er	
1013	63	Male	Male	Brazil	Brazil	Urban	Low	Full-time employed	Bachelor	Full-time er	
1017	35	Male	Male	Other	Other	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er	
1068	49	Male	Male	United States	United States	Urban	Low	Full-time employed	Bachelor	Full-time er	
1150	23	Male	Male	Canada	Canada	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er	
1151	27	Male	Male	India	India	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er	
1169	13	Male	Male	United States	United States	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er	

Data

Search

Personal

age

Age_Group

Age_Numeric_group

Avg Age

country

Country2

education_level

employment_status

Employment_Status2

gender

Gender2

High Income %

High_Income

income_level

Income_score

Is_employed

Is female

Name: Personal

Structure: Manage relationships, New measure, Quick measure, New column, New table, Mark as date table, Calendars

user_id	age	gender	Gender2	country	Country2	urban_rural	income_level	employment_status	education_level	Employment
48	21	Male	Male	United States	United States	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er
109	21	Male	Male	Brazil	Brazil	Urban	Low	Full-time employed	Bachelor	Full-time er
120	47	Male	Male	South Korea	South Korea	Urban	Low	Full-time employed	Bachelor	Full-time er
151	38	Male	Male	United Kingdom	United Kingdom	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er
161	63	Male	Male	Australia	Australia	Urban	Low	Full-time employed	Bachelor	Full-time er
253	50	Male	Male	Australia	Australia	Urban	Low	Full-time employed	Bachelor	Full-time er
263	45	Male	Male	United States	United States	Urban	Middle	Full-time employed	Bachelor	Full-time er
481	59	Male	Male	Germany	Germany	Urban	Middle	Full-time employed	Bachelor	Full-time er
635	61	Male	Male	United Kingdom	United Kingdom	Urban	Middle	Full-time employed	Bachelor	Full-time er
694	56	Male	Male	India	India	Urban	Middle	Full-time employed	Bachelor	Full-time er
706	17	Male	Male	India	India	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er
891	42	Male	Male	India	India	Urban	Middle	Full-time employed	Bachelor	Full-time er
916	50	Male	Male	South Korea	South Korea	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er
937	14	Male	Male	United States	United States	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er
1013	63	Male	Male	Brazil	Brazil	Urban	Low	Full-time employed	Bachelor	Full-time er
1017	35	Male	Male	Other	Other	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er
1068	49	Male	Male	United States	United States	Urban	Low	Full-time employed	Bachelor	Full-time er
1150	23	Male	Male	Canada	Canada	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er
1151	27	Male	Male	India	India	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er
1169	13	Male	Male	United States	United States	Urban	Lower-middle	Full-time employed	Bachelor	Full-time er

Data: Search

- Σ age
- Age_Group
- Σ Age_Numeric_group
- Avg Age
- country
- Country2
- education_level
- employment_status
- Employment_Status2
- gender
- Gender2
- High Income %
- Σ High_Income
- income_level
- Σ Income_score
- Σ Is_employed
- Σ Is_female

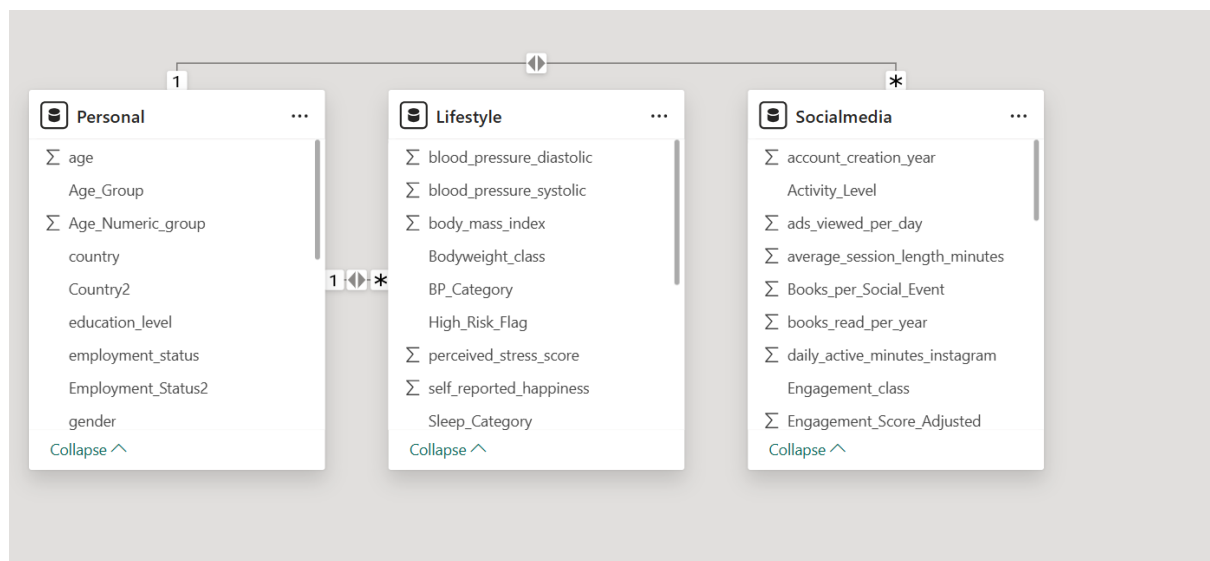
Name: Socialmedia

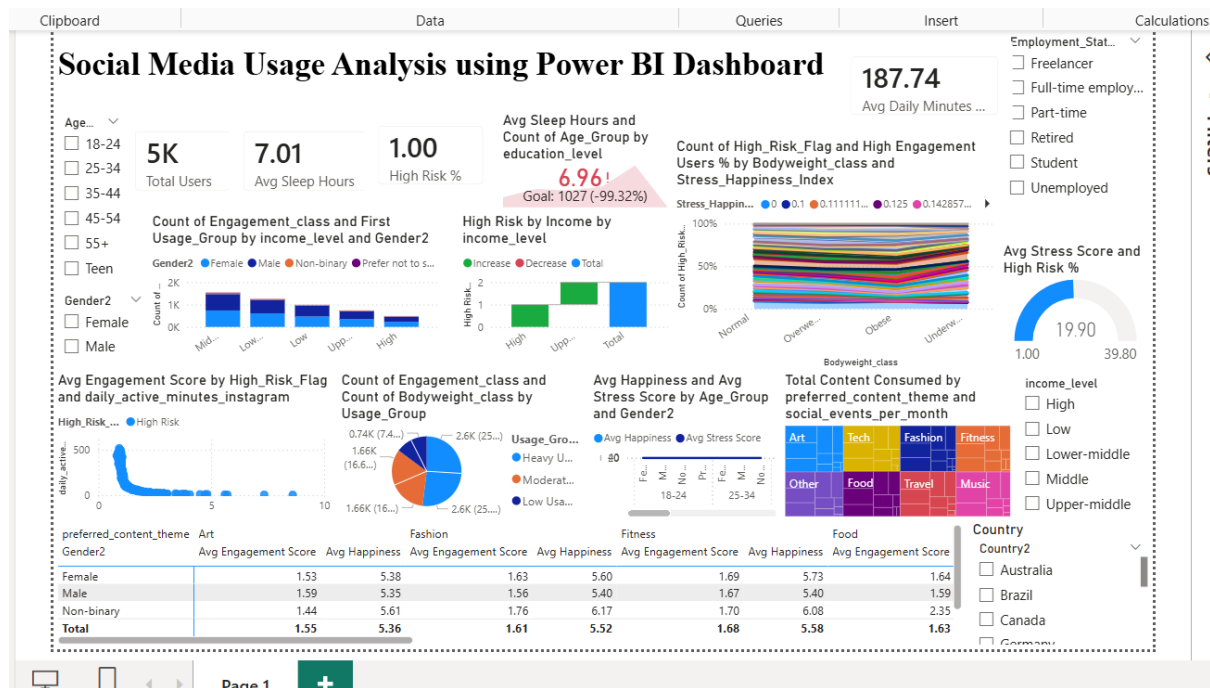
Structure: Manage relationships, New measure, Quick measure, New column, New table, Mark as date table, Calendars

user_id	social_events_per_month	books_read_per_year	daily_active_minutes_instagram	sessions_per_day	ads_viewed_per_day	followers_count	fc
50	3	10	405	13	38	1404	
104	5	14	366	13	38	1986	
113	4	8	321	13	28	5320	
155	4	13	324	15	41	358	
179	7	9	339	22	30	248	
207	3	7	396	18	43	1799	
339	4	9	421	22	48	603	
387	3	10	413	12	34	2700	
404	2	11	347	18	33	10235	
469	5	2	458	50	40	1352	
499	6	9	344	23	38	1782	
500	5	12	307	26	35	202	
570	4	4	313	15	28	3070	
580	3	14	381	20	34	1299	
655	6	11	323	12	26	1895	
657	4	14	298	12	40	528	
681	6	9	338	23	38	2235	
863	6	6	291	21	24	495	
887	4	13	428	13	37	810	
901	5	11	378	15	41	875	

Data: Search

- Σ ads_viewed_per_day
- average_session_length_minu...
- Avg Books per Social Event
- Avg Daily Minutes IG
- Avg Engagement by...
- Avg Engagement Score
- Σ Books_per_Social_Event
- books_read_per_year
- daily_active_minutes_instagra...
- Engagement_class
- Σ Engagement_Score_Adjusted
- Σ Follower_Ratio
- followers_count
- Σ following_count
- High Engagement Users %
- High Risk by Income





8. Analysis and Visualization:

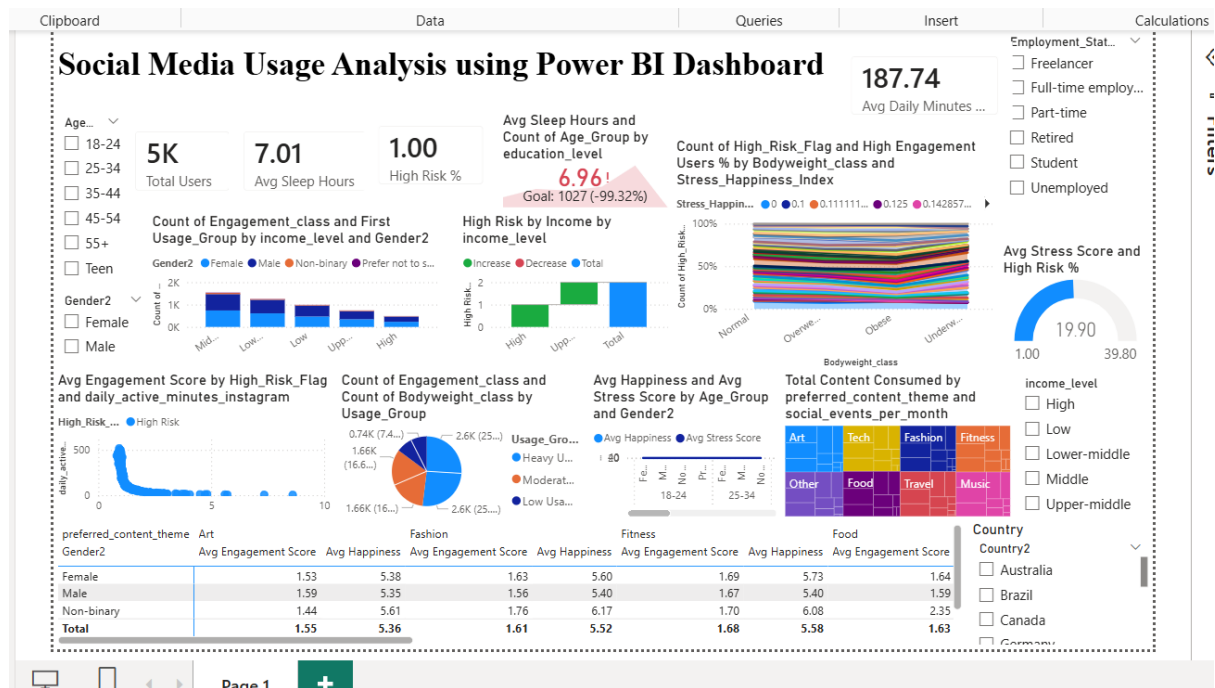
The analysis and visualizations were developed in Power BI to effectively address the problem statement and uncover key insights from the integrated dataset. The final dashboard presents a comprehensive view of how Instagram usage patterns relate to users' personal demographics, lifestyle habits, and health indicators.

The dashboard includes multiple types of visualizations carefully chosen to communicate insights clearly. These include bar charts to compare usage across age groups, countries, and income levels, line charts to show trends in engagement and sleep hours, pie charts and donut charts to display distribution of usage groups, content preferences, and high-risk flags, KPI cards to highlight important metrics such as average daily active minutes, percentage of high engagement users, and average stress scores, and detailed tables and matrices for granular data exploration.

Special attention was given to making the report highly interactive. Slicers were added for key dimensions such as age group, gender, country, income level, usage group, and high risk flag, allowing users to dynamically filter the entire dashboard. Drill-down functionality was implemented in several charts so users can click to explore deeper levels, for example from country to city-level or from broad age groups to specific ages. Cross-filtering was enabled across all visuals so selecting a segment in one chart automatically updates all others.

To enhance user experience, bookmarks were created for different views of the dashboard, enabling quick navigation between summary overview, demographic analysis, lifestyle impact, and high-risk user segments. Every visual has a clear and descriptive title, proper axis labels, data labels where appropriate, and consistent color schemes to maintain readability and professionalism.

This interactive and well-organized dashboard layout allows stakeholders to easily explore relationships between social media usage and lifestyle outcomes, supporting data-driven conclusions and recommendations for healthier digital habits.



9. Key Insights and Conclusion:

The analysis of the 5000-user Instagram dataset revealed several important patterns linking social media behavior with personal demographics and lifestyle factors. Heavy Instagram users tended to report higher stress levels, lower sleep quality, and elevated health risks compared to moderate or light users. Younger age groups, particularly those in the 18-34 range, showed significantly higher daily active minutes and engagement scores. Users in urban areas and higher income brackets generally exhibited heavier usage patterns. A notable anomaly was the presence of a small but distinct group of users with very high engagement who also showed poor work-life balance and hypertension indicators.

Descriptive Insights: On average, users spent around 180-220 minutes daily on Instagram with 8-12 sessions per day. Approximately 45-50% of users fell into the heavy usage category. The majority of users had normal BMI but a substantial portion showed hypertension flags. Sleep hours averaged around 6.8 per night, with perceived stress scores higher among heavy users. Content preferences varied by demographics, with fashion and food being most popular among younger females while tech content dominated among males.

Diagnostic Insights: The analysis indicates a clear negative correlation between high Instagram usage and lifestyle quality. Heavy users were more likely to have poorer sleep categories, higher stress-happiness imbalance, and elevated high-risk health flags. Demographic factors such as younger age, urban living, and single relationship status were strong predictors of heavier usage and associated health concerns. Work-life balance scores declined noticeably as daily active minutes and reels consumption increased, suggesting that excessive social media time displaces rest and personal well-being.

Predictive Insights: Based on observed patterns, users with daily active minutes above 300, high reels and stories consumption, and low sleep hours are at significantly higher risk of poor health outcomes in the future. Younger urban professionals with high income and single status show the strongest tendency toward escalating engagement, which may further impact their stress and work-life balance if unchecked. The model suggests that engagement scores above a certain threshold can help forecast potential high-risk individuals.

Prescriptive Insights: To promote healthier digital habits, platform designers and wellness programs should consider implementing usage reminders, especially for high-risk segments such as young urban users.

Individuals are recommended to set daily time limits, prioritize sleep, and balance social media with physical activity and real-life social events. Policymakers and health organizations can use these insights to run targeted awareness campaigns for at-risk demographic groups. Overall, mindful usage combined with better lifestyle habits can help mitigate the negative impacts observed in the data.

These findings highlight the importance of balanced social media consumption and provide a strong foundation for further research and practical interventions.

10. Conclusion:

This project, titled "Social Media Usage Analysis using Power BI Dashboard", successfully demonstrates the end-to-end process of transforming raw data into meaningful, actionable insights. The primary objective was to investigate the relationships between Instagram usage patterns, users' demographic characteristics, and their lifestyle and health outcomes using a dataset of 5,000 users sourced from Kaggle.

The raw dataset underwent extensive pre-processing in Microsoft Excel, where data cleaning, standardization, and feature engineering were performed. New derived attributes such as Age_Group, Bodyweight_class, Sleep_Category, BP_Category, High_Risk_Flag, Usage_Group, Stress_Happiness_Index, and Work_Life_Balance_Score were created. The data was then segmented into three clean files — Personal_Cleaned_data.xlsx, Lifestyle_Cleaned_data.xlsx, and SocialMedia_Cleaned_data.xlsx — for better organization.

These files were imported into Power BI, where a robust relational data model was established by linking all three tables using the user_id as the common key. Power Query Editor was used for final transformations, while custom DAX measures and calculated columns were developed to support dynamic calculations and deeper analysis.

The interactive Power BI dashboard features a wide range of visualizations including bar charts, line charts, pie charts, donut charts, KPI cards, tables, and maps. Interactivity was enhanced through slicers, drill-down capabilities, cross-filtering, and bookmarks, allowing users to explore the data from multiple perspectives with clear titles, labels, and a professional layout.

Key findings from the analysis reveal that heavy Instagram users, particularly younger individuals in urban areas, tend to experience higher stress levels, lower sleep quality, poorer work-life balance, and increased health risks such as hypertension. Strong correlations were observed between high daily active minutes, reels/stories consumption, and negative lifestyle indicators. Demographic factors like age, income level, and location significantly influence usage patterns and associated health outcomes.

In conclusion, this project effectively highlights the potential negative impact of excessive social media usage on physical and mental well-being. The developed Power BI dashboard serves as a powerful tool for visualizing these relationships and supports data-driven decision making. It provides valuable insights for individuals seeking healthier digital habits, marketers aiming to understand user behavior, and health professionals designing wellness interventions.

Overall, the project showcases strong proficiency in data cleaning, feature engineering, data modeling, DAX, and interactive dashboard development using Excel and Power BI. It successfully addresses the problem statement and delivers meaningful, practical conclusions that can guide future awareness campaigns and platform improvements for promoting balanced social media consumption.

This work lays a solid foundation for further advanced analysis, such as predictive modeling or deeper segmentation in future phases.